

IADI Decision Log

Indo-American Defence Innovation Index

A record of every meaningful decision made while building the index, what was tried, what was rejected, why I stopped where I did, and what I would revisit.

Version

1.1

Backtest Period

Jan 2022 – Apr 2026

Decisions Logged

15

Design Decisions

Each entry covers: what I tried · what I rejected and why · where I stopped · the tradeoff I accepted.

01 Why IADI Exists

I looked at existing ETFs and found none that captured both the US and Indian defence sectors simultaneously. That gap became the entire reason for building IADI, not as an academic exercise but as a product that fills a real market hole.

02 Universe Scope - India Only vs US Only vs Both

I initially thought to focus on either India or the US. After researching existing products, I realised no ETF covered both markets together. I expanded the scope to include both, which also made the index more resilient since the two defence sectors are driven by different macro forces.

03 Why Only US and India

These are the two markets I personally have access to invest in. This created a conscious bias, I could have included UK, France, Israel, or South Korea for a truly global index, but data access, currency complexity, and personal familiarity kept the scope to US and India.

04 Geography Split - Hard 60/40 vs Mathematical

I initially thought to enforce a hard 60% US / 40% India split. Instead I let the eligibility filters and market cap weighting determine the split organically. The result came out at 66% US / 34% India, close to my original intuition.

05 Universe Source

I first tried pulling the universe from the S&P 500 Wikipedia list filtered by GICS Aerospace & Defence sub-industry. This pulled in too many diversified industrials with minimal defence revenue. I switched to NSE EQUITY_L.csv with keyword matching on company names for the India side, and a manually curated list of US pure-plays.

06 Revenue Threshold - 20% vs 40% vs 33%

I considered a 20% and 40% revenue threshold which made it too wide or too narrow. I settled on 33% (one-third) because it kept the stock range between what the 20% and 40% thresholds produced. This was a judgment call, not a formally optimised number.

07 Manual Overrides - Force Includes and Excludes

The design was fully rules-based but the keyword filter had two failure modes where it let in the wrong companies and kept out the right ones. So I added all legitimate defence companies whose yfinance descriptions were too sparse or generic to hit the keyword threshold despite being well-known

government contractors while excluding big companies who do not service much revenue from defence.

08 Currency Conversion - USD to INR

Since IADI is designed as an India-listed ETF, all values are in INR. US stock prices were converted daily using the USD/INR exchange rate from yfinance. Where exchange rate data was missing, I forward-filled up to 5 days, which is standard since rates do not move materially over short gaps.

09 Reconstitution Frequency - Annual vs Semi-Annual

I started with annual reconstitution. Testing semi-annual reconstitution showed better Sharpe, better returns, and lower maximum drawdown. I did not test quarterly because it would overlap with the rebalance schedule and add operational complexity without clear benefit.

10 Fixed vs Dynamic Constituents

I initially planned for a fixed number of constituents. I shifted to a dynamic approach where the number is determined by how many stocks pass the eligibility filters at each reconstitution. This is more realistic and avoids forcing inclusions just to hit a target count.

11 Weight Cap - 5% vs 10%

I started with a 5% cap per stock. Testing at 10% produced better results. I did not push further because beyond 10% the index starts showing meaningful single-stock concentration risk, which defeats the purpose of a diversified index.

12 Weighting Method

I started with inverse volatility weighting as the baseline. I tested six methods in total namely inverse volatility, equal weight, market cap, minimum variance, maximum diversification, and risk parity. Market cap weighting produced the best Sharpe ratio and was also the most transparent and operationally simple.

13 Benchmark - India Leg Reconstructed Manually

I needed an India defence benchmark to compare against. The Nifty India Defence Index exists but its historical data is not available through any free source. Rather than drop the India leg entirely or substitute Nifty 50, which would be the wrong theme, I reconstructed the index manually using semi-annual constituent snapshots sourced from NSE factsheets and historical ETF disclosures. I used the 6-month average price as a proxy for free-float market cap since actual free-float factors were not freely available. This is an acknowledged approximation and would be replaced with official index data in a production setting.

14 Transaction Costs - Excluded Deliberately

I initially planned to include transaction costs in the backtest. I dropped this because US and Indian markets have very different cost structures like STT, SEBI charges, bid-ask spreads, brokerage and there was no free reference data to normalise across both markets fairly. Including arbitrary assumed costs would make the backtest less credible, not more. I instead disclose turnover at every rebalance event so costs can be estimated separately by anyone reviewing this.

Key Outcomes

- Final index level: ₹6,890 from a ₹1,000 base (January 2022 – April 2026)
- Best weighting method: Market Cap - Sharpe ratio = 2.340
- Constituent count grew organically: 28 stocks (2022) → 34 stocks (2026)
- Average quarterly turnover: 6.2% - efficient and implementable
- Geography split: 66% US / 34% India - emerged naturally from filters